

11/01/2023

MODULE- 05

MAXWELL'S EQUATIONS.

① Faraday's law:

The magnitude of Induced emf in the circuit is equal to the rate of change of magnetic flux through it and its direction is opposite to flux.

$$\text{emf or } \mathcal{E} = -\frac{d\Phi}{dt}$$

If N no. of turns are there in coil, then induced emf is given as

$$\mathcal{E} = -N \frac{d\Phi}{dt}$$

②

Maxwell's equation from Faraday's law.

The induced emf is a scalar quantity and is measured in volts.

From Faraday's law the induced emf in terms of electric field intensity is given by,

$$\mathcal{E} = \oint_C \mathbf{E} \cdot d\mathbf{l} \quad \text{--- (1)}$$

The induced emf according to Faraday's law is

$$\mathcal{E} = -\frac{d\Phi}{dt} \quad \text{--- (1a)}$$

From the definition of magnetic flux density $\mathbf{B}$

$$\mathbf{B} = \frac{d\Phi}{ds} \text{ Tm}^2$$

The total magnetic flux passing through a specified area ds is given by,

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} \quad \text{--- (2)}$$

From Faraday's law,

Substitute equation (2) in (1a)

$$\mathcal{E} = -\frac{d}{dt} \left(\int_S \mathbf{B} \cdot d\mathbf{s} \right) \quad \text{--- (2a)}$$

Equating (1) and (2a)

$$\int_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \left(\int_S \mathbf{B} \cdot d\mathbf{s} \right) \rightarrow (3)$$

therefore,

the above equation (3) can be rewritten as

$$\int_C \mathbf{E} \cdot d\mathbf{l} = \int_S \frac{d}{dt} (\mathbf{B}) \cdot d\mathbf{s} \rightarrow (4)$$

$\rightarrow$ eqn (4) is maxwells equation in integral form.

• Maxwells equation from Faradays law in point form

Using the stokes theorem,

$$\oint_C \mathbf{A} \cdot d\mathbf{l} = \int_S \nabla \times \mathbf{A} \cdot d\mathbf{s}$$

Considering the LHS of expression (3), we get

$$\int_C \mathbf{E} \cdot d\mathbf{l} = \oint_S \nabla \times \mathbf{E} \cdot d\mathbf{s} \rightarrow (5)$$

Since,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \rightarrow (6)$$

The eqn (6) is called the maxwells equation in point form.

Problem:

1. If the electric field intensity in free space is given in rectangular co-ordinates as $E = E_m \sin(\alpha x) \sin(\omega t - \beta z) \hat{y} \text{ V/m}$. Find magnetic field intensity using Faraday's law.

Ans: $\nabla \times E = -\frac{\partial B}{\partial t} \quad (6)$

$$\nabla \times E = \begin{bmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{bmatrix} = -\frac{\partial B}{\partial t}$$

$$B = \mu H$$

$$B = \mu_0 \mu_r H$$

$$\mu_r = 1$$

$$B = \mu_0 H$$

$$= \begin{bmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & 0 & \frac{\partial}{\partial z} \\ 0 & E_m \sin(\alpha x) \sin(\omega t - \beta z) & 0 \end{bmatrix} = -\mu_0 \frac{\partial H}{\partial t}$$

$$= \hat{a}_x \left[0 - \frac{\partial}{\partial z} (E_m \sin(\alpha x) \sin(\omega t - \beta z)) \right] - 0 + \hat{a}_z \left[\frac{\partial}{\partial x} (E_m \sin(\alpha x) \sin(\omega t - \beta z)) \right]$$

$$= (-E_m \sin(\alpha x) \beta \cos(\omega t - \beta z)) \hat{a}_x + \hat{a}_z (E_m \cos(\alpha x) \sin(\omega t - \beta z))$$

$$-\frac{\partial H}{\partial t} = \frac{-E_m \beta}{\mu_0} \sin(\alpha x) \cos(\omega t - \beta z) \hat{a}_x + \frac{E_m \alpha}{\mu_0} \cos(\alpha x) \sin(\omega t - \beta z) \hat{a}_z$$

$$H = \int \frac{\partial H}{\partial t} dt = \frac{-E_m \beta}{\mu_0 \omega} \sin(\alpha x) \sin(\omega t - \beta z) \hat{a}_x + \frac{E_m \alpha}{\mu_0 \omega} \cos(\alpha x) \cos(\omega t - \beta z) \hat{a}_z \text{ A/m}$$

Q. For a given electric field, in free space.
Find D , B and H in $E = E_m \sin(\omega t - \beta z) a_y$ V/m.

Ans: $D = \epsilon_0 E$

$$D = (8.854 \times 10^{-12}) (E_m \sin(\omega t - \beta z) a_y) \quad \epsilon_0 = 8.854 \times 10^{-12}$$

$B =$

$$\nabla \times E = -\frac{\partial B}{\partial t} = -\frac{\partial \mu_0 H}{\partial t} \Rightarrow \mu_0 \frac{\partial H}{\partial t}$$

$$\begin{vmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & 0 & \frac{\partial}{\partial z} \\ 0 & E_m \sin(\omega t - \beta z) & 0 \end{vmatrix} = -\frac{\partial B}{\partial t}$$

$$-\frac{\partial B}{\partial t} = a_x \left[0 - \frac{\partial}{\partial z} (E_m \sin(\omega t - \beta z)) \right] - a_y(0) + a_z(0).$$

$$-\frac{\partial B}{\partial t} = E_m \beta \cos(\omega t - \beta z) a_x$$

$$B = -\frac{E_m \beta}{\omega} \sin(\omega t - \beta z) a_x \quad \text{Tesla.}$$

$$B = \mu_0 H$$

$$\mu_0 H = -\frac{E_m \beta}{\omega} \sin(\omega t - \beta z) a_x$$

$$H = -\frac{E_m \beta}{\omega \mu_0} \sin(\omega t - \beta z) a_x \quad \text{A/m}$$

3. with usual notations derive maxwells equation in point form as derived from faradays law. Hence show that the electric field $E = 2xz^3 a_x + 4x a_y$ V/m cannot arise from static distribution of charges.

• Note: when $\nabla \times E = 0$. It indicates that the static distribution of charges rises the electric field. On other hand,

$\nabla \times E \neq 0$, it indicates that it will not rise the electric field.

Ans $\nabla \times E = \begin{bmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & 0 & 0 \\ 2xz^3 & 4x & 0 \end{bmatrix}$

$$= a_x(0) - a_y(0) + a_z\left(\frac{\partial}{\partial x}(4x)\right)$$

$$\boxed{\nabla \times E = 4 a_z}$$

$\therefore \nabla \times E \neq 0.$

It will not rise electric field.

1. For given electric field in free space, find D, H and B , $E = 8^2 a_x + 28 a_y$ V/m.

Ans $D = \epsilon_0 E$

$$\boxed{D = 8.854 \times 10^{-12} (8^2 a_x + 28 a_y)}$$

$$\nabla \times E = \begin{bmatrix} a_x & 8 a_y & a_z \\ \frac{1}{8} \frac{\partial}{\partial t} & 0 & 0 \\ 8^2 & 28^2 & 0 \end{bmatrix} = -\frac{\partial B}{\partial t}$$

$$= a s(t) - s a(t) + a z \left(\frac{\partial}{\partial s} (2s^2) - 0 \right) \times \frac{1}{s}$$

$$= (2 \cdot 2s) a z \times \frac{1}{s}$$

$$= \frac{4s a z}{s}$$

$$\boxed{\nabla \times E = 4 a z}$$

$$4 a z = - \frac{\partial B}{\partial t}$$

$$B = - \int 4 a z \cdot dt$$

$$\boxed{B = - 4 a z t}$$

$$H = \frac{B}{\mu_0} = - \frac{4 t}{\mu_0} a z$$

$$\boxed{H = - \frac{4 t}{\mu_0} a z \text{ A/m}^2}$$

Inconsistency of Ampere's law / Modified ampere's law.

From ampere's law,

$$\nabla \times H = J \text{ A/m}^2 \longrightarrow (1)$$

Take divergence on Both Sides.

$$\nabla \cdot (\nabla \times H) = \nabla \cdot J$$

$$0 = \nabla \cdot J \longrightarrow (2) \quad (\text{The divergence on curl is } \nabla \times H = 0)$$

The eqn (2) shows that,

$\nabla \cdot J$ is not consistent with Continuity of Current equation.

$$\text{Since, } \nabla \cdot J = - \frac{\partial \rho_v}{\partial t}$$

∴ It is observed that the ampere's law is inconsistent and it needs some modification.
 Let us add some vector 'Q' to eqn (1)

$$\nabla \times H = J + Q \longrightarrow (3)$$

perform divergence on eqn (3) on both sides.

$$\nabla \cdot (\nabla \times H) = \nabla \cdot J + \nabla \cdot Q$$

$$0 = \nabla \cdot J + \nabla \cdot Q$$

$$\nabla \cdot J = -\nabla \cdot Q \longrightarrow (4)$$

Since, $\nabla \cdot J = -\frac{\partial \rho_v}{\partial t}$

the eqn (4) can be re-written as

$$-\nabla \cdot Q = -\frac{\partial \rho_v}{\partial t}$$

$$\boxed{\nabla Q = \frac{\partial \rho_v}{\partial t}} \longrightarrow (5)$$

From point form of Gauss law,

$$\nabla \cdot D = \rho_v \longrightarrow (6)$$

Differentiate eqn (6) w.r.t. dt.

$$\nabla \cdot \frac{\partial D}{\partial t} = \frac{\partial \rho_v}{\partial t} \longrightarrow (7)$$

Using equation (7) in (5)

$$\nabla \cdot Q = \nabla \cdot \frac{\partial D}{\partial t} \longrightarrow (8)$$

$$\therefore \boxed{Q = \frac{\partial D}{\partial t}}$$

∴ ampere's law is modified to

$$\boxed{\nabla \times H = J + \frac{\partial D}{\partial t}} \longrightarrow (9)$$

Point form of modified ampere circuit law

Here $\frac{\partial D}{\partial t}$ is called displacement current density & J is called conduction current density.

$$\therefore J_{\text{total}} = J_c + J_D \longrightarrow$$

Integrating equation (9) on B.S.

$$\int_S (\nabla \times H) \cdot ds = \int_S \left(J + \frac{\partial D}{\partial t} \right) \cdot ds$$

$$\int_S (\nabla \times H) \cdot ds = \int_S (J_c + J_D) \cdot ds$$

Applying the Stokes theorem,

$$\int_L H \cdot dl = \int_S (\nabla \times H) \cdot ds = I \text{ amps.}$$

$$\therefore \int_L H \cdot dl = \int_S (J_c + J_D) \cdot ds = I$$

$$= \int_S J_c \cdot ds + \int_S J_D \cdot ds = I$$

$$\int_L H \cdot dl = \sqrt{J_c^2 + J_D^2} = I$$

• Conduction Current (J_c).

The Conduction current density (J_c) in terms of electric field intensity (E) is given as $J_c = \sigma E \text{ A/m}^2 \longrightarrow (1)$

Since, the Conduction Current J_c can be represented as

$$i_c = J_c A$$

$$\therefore |i_c| = |J_c| A$$

$$|J_c| A = \int J_c \cdot ds \text{ amps}$$

• Displacement Current density (J_D)

$$J_D = \frac{\partial D}{\partial t} \text{ A/m}^2 \longrightarrow (2)$$

$$i_D = |J_D| \text{ A, amps}$$

∴ The total Current density,

$$J = J_C + J_D$$

$$J = \sigma E + \frac{\partial D}{\partial t} \text{ A/m}^2 \longrightarrow (3)$$

The fourier transform of

$$\frac{\partial}{\partial t} \xrightarrow{FT} j\omega$$

∴ eqn (3) $\Rightarrow$

$$J = \sigma E + j\omega E$$

$$\boxed{J = E(\sigma + j\omega \epsilon)}$$

~~Continuity~~ Current equation from
maxwells equation.

• Continuity Current equation from maxwells equation

a) with usual notation derive the differential forms of Continuity Current equation from maxwells equation.

$$\rightarrow \nabla \cdot J = -\frac{\partial \rho_v}{\partial t} \longrightarrow (1)$$

from ampere's modified circuital law,

$$\nabla \times H = J_C + \frac{\partial D}{\partial t} \text{ A/m}^2 \longrightarrow (2)$$

Take divergence on eqn (2) from B.S.

$$\nabla \cdot (\nabla \times H) = \nabla \cdot J_C + \nabla \cdot \frac{\partial D}{\partial t}$$

$$0 = \nabla \cdot \mathbf{J}_c + \nabla \cdot \frac{\partial \mathbf{D}}{\partial t}$$

$$\nabla \cdot \frac{\partial \mathbf{D}}{\partial t} = -\nabla \cdot \mathbf{J}_c \longrightarrow (3)$$

eqn (3) is called, point form of Gauss law:

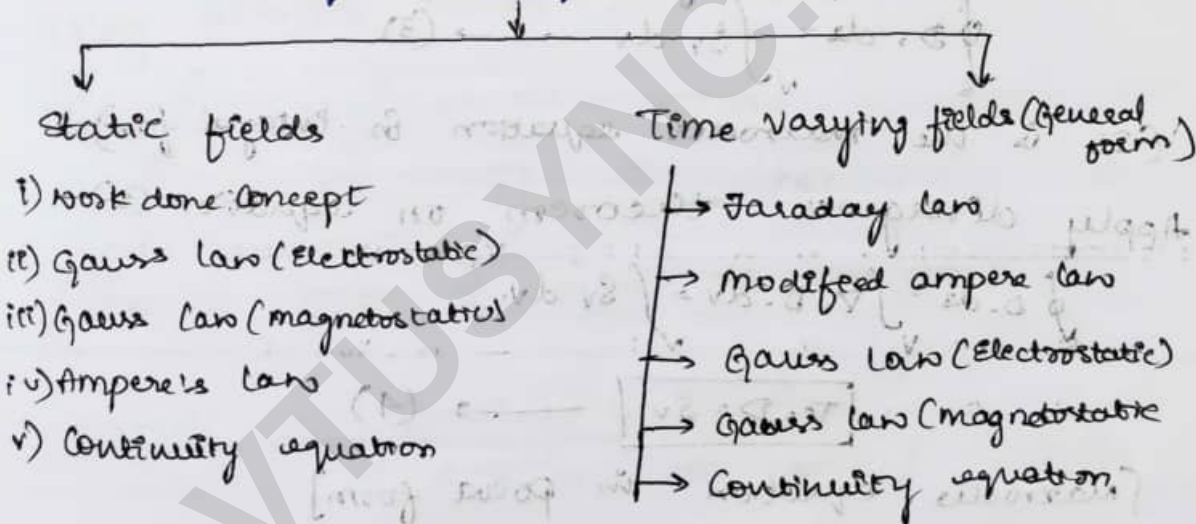
$$\frac{\partial}{\partial t} \nabla \cdot \mathbf{D} = -\nabla \cdot \mathbf{J}_c.$$

Since $\nabla \cdot \mathbf{D} = \rho_v$

the above equation is re-written as,

$$\boxed{\frac{\partial \rho_v}{\partial t} = -\nabla \cdot \mathbf{J}_c} \rightarrow \text{differential form of Maxwell's equation.}$$

Classification of Maxwell's Equation



Derivation of Maxwell's equation for static fields.

i) Based on work-done concept.

The work done required to move a point charge of unit coulombs over a closed path is equal to zero.

$$\therefore \oint \mathbf{E} \cdot d\mathbf{l} = 0 \longrightarrow (1)$$

using Stokes theorem.

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = \int_S \nabla \times \mathbf{E} \cdot d\mathbf{s} = 0$$

Here, $\nabla \times \mathbf{E} = 0$ when $d\mathbf{s} \neq 0$.

$\therefore \nabla \times \mathbf{E} = 0 \longrightarrow (2)$ (Maxwell's equation in point form)

2) Based on Gauss law [Electro-static]

The total flux coming out of any closed surface is equal to the total charge enclosed by that surface.

$$\therefore \oint_S \mathbf{D} \cdot d\mathbf{s} = Q_{\text{enclosed}} = \int_V \rho_v dv$$

$\therefore$ The surface integration,

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \int_V \rho_v dv \longrightarrow (3)$$

(It is the Maxwell's equation in integral form)

Apply divergence theorem on equation (3)

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \int_V \nabla \cdot \mathbf{D} \cdot dv = \int_V \rho_v dv$$

$$\therefore \boxed{\nabla \cdot \mathbf{D} = \rho_v} \longrightarrow (4)$$

(Maxwell's equation in point form)

3) Based on Gauss law [Magnetostatic]

The magnetic flux lines form closed path i.e. The total outgoing flux is equal to the incoming magnetic flux.

$$\therefore \oint_S \mathbf{B} \cdot d\mathbf{s} = 0 \longrightarrow (5)$$

(Maxwell's equation in integral form)

Apply divergence theorem on equation (5)

$$\oint_S \mathbf{B} \cdot d\mathbf{s} = \int_V \nabla \cdot \mathbf{B} \cdot dv = 0$$

$$\therefore \boxed{\nabla \cdot B = 0} \longrightarrow (6)$$

[Maxwell's equation in point form]

4) Ampere's Circuital Law:

The line integral of H around a closed path is equal to the current enclosed.

$$\oint H \cdot dl = I \text{ amps} = \int_S J \cdot ds$$

$\therefore$ The line integration,

$$\oint H \cdot dl = \int_S J \cdot ds \text{ amps } (7)$$

(Maxwell's equation in integral form)

Using Stokes theorem,

$$\text{i.e. } \oint H \cdot dl = \int_S \nabla \times H \cdot ds = \int_S J \cdot ds$$

$$\therefore \boxed{\nabla \times H = J} \longrightarrow (8)$$

(point form of Maxwell's equation)

5) Based on Continuity equation.

From the concept of current density

$$\text{i.e. } J = \frac{dI}{ds} \text{ amp/m}^2$$

$$\therefore I = \oint J \cdot ds = \frac{dQ}{dt}$$

Since for static field,

$$Q = \text{const}$$

$$\frac{dQ}{dt} = 0$$

$$\therefore \oint J \cdot ds = 0 \longrightarrow (9)$$

(integral form) of ~~Maxwell's~~

Apply divergence theorem on (9)

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = \int_V \nabla \cdot \mathbf{J} dv = 0$$

$$\therefore \nabla \cdot \mathbf{J} = 0 \longrightarrow (10)$$

(point form)

In the above derivation from eq (1) to eq (10) are called maxwells equations for static field.

In the above derivations, the eq (1), (3), (5) (7) and (9) are called maxwells equation in integral form. and eq (2), (4), (6), (8) (10) are maxwells equation in point form.

List maxwells equation for static fields.

S/NO	Integral	Point form
1.	$\oint_L \mathbf{E} \cdot d\mathbf{l} = 0$	$\nabla \times \mathbf{E} = 0$
2.	$\oint_S \mathbf{D} \cdot d\mathbf{s} = \int_V \rho_v dv$	$\nabla \cdot \mathbf{D} = \rho_v$
3.	$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$	$\nabla \cdot \mathbf{B} = 0$
4.	$\oint_L \mathbf{H} \cdot d\mathbf{l} = \oint_S \mathbf{J} \cdot d\mathbf{s}$	$\nabla \times \mathbf{H} = \mathbf{J}$
5.	$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0$	$\nabla \cdot \mathbf{J} = 0$

6. Maxwell's equation of time varying fields

1. Based on Modified amperes law.

The total current, $I = \int H \cdot dl = \int (J_c + J_D) \cdot ds$

Since $J_c = \sigma E \text{ A/m}^2$, $J_D = \frac{\partial D}{\partial t} \text{ A/m}^2$

$$\oint H \cdot dl = \int (J_c + \frac{\partial D}{\partial t}) \cdot ds \rightarrow (1)$$

(integral form of Maxwell's)

Applying divergence theorem on eq (1)

i.e.

$$\oint H \cdot dl = \int \nabla \times H \cdot ds$$

$$\int \nabla \times H \cdot ds = \int (J_c + \frac{\partial D}{\partial t}) \cdot ds$$

$$\nabla \times H = J_c + \frac{\partial D}{\partial t} \rightarrow (2)$$

(point form of Maxwell's)

2. Based on Faradays law.

Refer derivation from starting of mod - 5
i.e. Maxwell's equation from Faraday's law)

3. Based on Gauss law (Electrostatics)

The total electric flux crossing the closed surface is equal to the total charge enclosed by the surface.

$$\oint D \cdot ds = \int \rho_v \cdot dv \rightarrow (3)$$

(Maxwell's eqn in integral form)

Apply divergence theorem on eq (3)

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \int_V \nabla \cdot \mathbf{D} dv = \int_V \rho_v dv$$

$$\therefore \boxed{\nabla \cdot \mathbf{D} = \rho_v} \rightarrow (4)$$

(point form of maxwells)

4. Based on Gauss (or) Magnetostatic]

In case of magnetic field to total
Outgoing flux is equal to incoming flux

$$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0 \rightarrow (5)$$

(Integral form)

Apply, divergence theorem on (5)

$$\oint_S \mathbf{B} \cdot d\mathbf{s} = \int_V \nabla \cdot \mathbf{B} dv = 0$$

$$\nabla \cdot \mathbf{B} = 0 \rightarrow (6) \text{ (point form)}$$

5. Based on Continuity equation

The Continuity Current equation from
the law of Conservation of charges

$$I = \oint_S \mathbf{J} \cdot d\mathbf{s} = -\frac{dQ}{dt} = -\int_V \frac{\partial \rho_v}{\partial t} dv$$

Apply divergence,

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = -\int_V \frac{\partial \rho_v}{\partial t} dv = \int_V \nabla \cdot \mathbf{J} dv$$

$$\boxed{\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}} \rightarrow (7) \text{ (point form)}$$

List the Maxwell's equation for time Varying field (General)

Sl No.	Integral	Point form.
1.	$\oint_C H \cdot dl = \oint_S (J_c + \frac{\partial D}{\partial t}) \cdot d\mathbf{s}$	$\nabla \times H = J_c + \frac{\partial D}{\partial t}$
2.	$\oint_C E \cdot dl = - \frac{d}{dt} \oint_S B \cdot d\mathbf{s}$	$\nabla \times E = - \frac{\partial B}{\partial t}$
3.	$\oint_S D \cdot d\mathbf{s} = \int_V \rho_v dv$	$\nabla \cdot D = \rho_v$
4.	$\oint_S B \cdot d\mathbf{s} = 0$	$\nabla \cdot B = 0$
5.	$\oint_S J \cdot d\mathbf{s} = - \int_V \frac{\partial \rho_v}{\partial t} dv$	$\nabla \cdot J = - \frac{\partial \rho_v}{\partial t}$

Module-05

Chapter-2

Plane waves

- Derive the wave equation for free space (22)

Derive uniform plane wave equation from Maxwell's equation representing E and H .

For free space, the volume charge density $\rho_v = 0$, $\rho = 0$, and $J_c/J = 0$.

and $\mu = \mu_0$.

From the Maxwell's equation.

$$\nabla \times H = J_c + \frac{\partial D}{\partial t} \text{ A/m}^2.$$

$$\nabla \times E = 0 + \frac{\partial}{\partial t} \epsilon_0 E \text{ A/m}^2.$$

$$\nabla \times H = \epsilon_0 \frac{\partial E}{\partial t} \quad (\text{Maxwell's equations for free space}).$$

From Maxwell equation Based on Faraday's law

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

$$\nabla \times E = -\frac{\partial \mu_0 H}{\partial t}$$

$$\boxed{\nabla \times E = -\mu_0 \frac{\partial H}{\partial t}}$$

$\therefore$ The Maxwell's equation for free space are

$$\nabla \times H = \epsilon_0 \frac{\partial E}{\partial t} \longrightarrow (1)$$

$$\nabla \times E = -\mu_0 \frac{\partial H}{\partial t} \longrightarrow (2)$$

perform curl on equation (2),

$$\nabla \times \nabla \times \mathbf{E} = -\mu_0 \frac{\partial (\nabla \times \mathbf{H})}{\partial t}$$

from Vector Identity,

$$\nabla \times \nabla \times \mathbf{A} = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}.$$

$$\nabla (\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\mu_0 \frac{\partial (\nabla \times \mathbf{H})}{\partial t} \rightarrow (3).$$

using the poissone equation

$$\nabla \cdot \mathbf{D} = \rho_v \rightarrow (4)$$

$$\nabla \cdot \epsilon_0 \mathbf{E} = \rho_v$$

$$\epsilon_0 \nabla \cdot \mathbf{E} = \rho_v$$

$$\nabla \cdot \mathbf{E} = \frac{\rho_v}{\epsilon_0} \rightarrow (4a)$$

Apply equation (4a) in (3)

$$\nabla \cdot \left(\frac{\rho_v}{\epsilon_0} \right) - \nabla^2 \mathbf{E} = -\mu_0 \frac{\partial (\nabla \times \mathbf{H})}{\partial t} \rightarrow (5)$$

Substitute $\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$

Substitute in eqn (5)

$$\nabla \cdot \left(\frac{\rho_v}{\epsilon_0} \right) - \nabla^2 \mathbf{E} = -\mu_0 \frac{\partial}{\partial t} \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right)$$

$$\nabla \cdot \left(\frac{\rho_v}{\epsilon_0} \right) - \nabla^2 \mathbf{E} = -\mu_0 \frac{\partial}{\partial t} \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$

$$\nabla \cdot \frac{\rho_v}{\epsilon_0} + \mu_0 \frac{\partial}{\partial t} \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) = \nabla^2 \mathbf{E}$$

$$\nabla \cdot \frac{\rho_v}{\epsilon_0} + \mu_0 \frac{\partial \mathbf{J}}{\partial t} + \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = \nabla^2 \mathbf{E}$$

$$\nabla \cdot \frac{\rho_v}{\epsilon_0} + \mu_0 \frac{\partial \mathbf{J}}{\partial t} = \nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \rightarrow (6)$$

eq(6) represents plane wave in E form

Similarly,

the plane wave equation in H. form can be derived as follows,

From eqn (1), general form,

$$\nabla \times H = J + \frac{\partial D}{\partial t}$$

$$\nabla \times H = J + \epsilon_0 \frac{\partial E}{\partial t} \rightarrow (2)$$

Take curl on both sides on (2):

$$\nabla \times \nabla \times H = \nabla \times J + \epsilon_0 \frac{\partial \nabla \times E}{\partial t}$$

$$\nabla(\nabla \cdot H) - \nabla^2 H = \nabla \times J + \epsilon_0 \frac{\partial (\nabla \times E)}{\partial t} \rightarrow (3)$$

As per maxwells equations $\nabla \cdot B = 0$.

$$\therefore \nabla \cdot H = 0$$

$$\therefore 0 - \nabla^2 H = \nabla \times J + \epsilon_0 \frac{\partial (\nabla \times E)}{\partial t} \rightarrow (4)$$

Substitute (2) in (4)

$$-\nabla^2 H = \nabla \times J + \epsilon_0 \frac{\partial (-\mu_0 \frac{\partial H}{\partial t})}{\partial t}$$

$$-\nabla^2 H = \nabla \times J - \epsilon_0 \mu_0 \frac{\partial^2 H}{\partial t^2}$$

$$\epsilon_0 \mu_0 \frac{\partial^2 H}{\partial t^2} - \nabla^2 H = \nabla \times J$$

Since for free space $J=0$.

$$\epsilon_0 \mu_0 \frac{\partial^2 H}{\partial t^2} - \nabla^2 H = 0 \rightarrow (10)$$

eq(10) represents plane wave equation in H form.

• Solution for plane wave equation.

$$\nabla^2 E - \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2} = 0 \rightarrow (1)$$

$$E = E_0 e^{j\omega t}$$

Substitute E in eq (1).

$$\nabla^2 E - \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} (E_0 e^{j\omega t}) = 0$$

$$\nabla^2 E - \mu_0 \epsilon_0 (J^2 \omega^2 E_0 e^{j\omega t}) = 0 \rightarrow (2)$$

eqn (2) is called Helmholtz equation and solution to this equation is

$$E = E_m \cos(\omega t - \beta z) + E_m \cos(\omega t + \beta z) \text{ V/m.}$$

Similarly,

the solution for plane wave equation

$$H = H_m \cos(\omega t - \beta z) - H_m \cos(\omega t + \beta z) \text{ A/m,}$$